

Shumaila Ambreen

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Education

Kansas State University, Manhattan, KS

Expected June 2028

Ph.D. in Mathematics

Advisor: Dinh-Liem Nguyen

Research Interests: Inverse scattering problems, elastic waves, periodic media

Quaid-e-Azam University, Islamabad, Pakistan

March 2013

M.Phil. in Mathematical Sciences (Pure Mathematics)

Thesis: A Study of Multiplicatively Conjugate Loops and Wielandt Subloops

Advisor: Muhammed Asif

Awards

- I-Center Scholarship, Spring 2026
- Dr. William L. Stamey Mathematics Scholarship, Spring 2026
- Friends of Mathematics Scholarship, Spring 2026
- Dr. William L. Stamey Mathematics Scholarship, Fall 2025

Publications

Published

- **S. Ambreen**, A. Ali. *A Study of Multiplicatively Conjugate Loops and Wielandt Subloops*, Quasigroups and Related Systems, 22 (2014), 1–8.

In preparation

- **S. Ambreen**, D-L. Nguyen and N. H. Nguyen. A Fast Imaging Function for Inverse Elastic Scattering by Penetrable Periodic Structures
- **S. Ambreen**, D.L. Nguyen, N. Nguyen, Aravindh K. Ravi, Cat-Su Tran. Resolution Analysis of Imaging Function under Weak Scattering Assumptions, 2026.

Talks and Presentations

Contributed Talks

- *A new imaging function for inverse scattering problems in elastic periodic media*
15th Ohio River Analysis Meeting, University of Kentucky Spring 2026

- *A new imaging function for inverse scattering problems in elastic periodic media*
SIAM Central States Section Meeting, University of Arkansas Fall 2025

Poster Presentation

- *A new imaging function for inverse scattering problems in elastic periodic media*
SIAM Central States Section Meeting, UMKC Spring 2025

Teaching Experience

- Calculus II (221) (Instructor and coordinator) Summer 2026
- Calculus III (222) (Recitation) Spring 2026
- Mathematical Modeling Seminar Spring 2026
- Elem Differential Eq (340) (Recitation) Fall 2025
- Calculus I, Trigonometry Geometry (Grader) Summer 2025
- Studio Algebra (100) (Recitation), Pathway (Instructor) Spring 2025
- Mathematical Modeling Seminar Spring 2025
- Top/College Algebra Workshop (199) (Instructor) Fall 2024
- Pathways Summer 2024
- Calculus II (221) (Recitation) Spring 2024
- Traditional Algebra (100) (Recitation) Fall 2023

Mentoring

- Undergraduate Research Mentor (Soleil Disney) Fall 2026
Topic: Mathematical modeling of reservoir dredging
- Undergraduate Research Mentor (Anna Starbuck) Spring 2026
Topic: Reconstruction of small sources from Cauchy data
- Directed Reading Program (Anna Starbuck) Fall 2025
Topic: Deep Learning

Service

- President, Association for Women in Mathematics (K-State) Fall 2025-Present
- Sonia Kovalevsky Day Organizer (Middle School Outreach) 2025–2026
- Graduate Recruitment Volunteer 2025–2026
- Graduate Program Information Session Volunteer Fall 2025

Skills

Technical: MATLAB, Python (PyTorch, TensorFlow, NumPy, SciPy, Pandas), GitHub

Languages: English (Fluent), Urdu (Native), Punjabi